Geometric DFM — review pack

12 findings across 6 real parts, for T / F / N marking — generated 2026-08-07

What this is. CostVision now runs geometric DFM rules directly against the measured CAD — not against the costing result, and with no AI involved. Every finding below names the B-rep faces that triggered it, the measured value, the threshold, and the published source of that threshold. The parts are the six with independent manual bottom-up costs, so they are the only ones whose right answer is known from outside the tool.

What we need from you. A mark against each finding. The rules have been validated by unit tests and by my own judgement, which proves the arithmetic is right and says nothing about whether the findings are *true*. Nobody with a manufacturing floor behind them has looked at one yet. That is the gap this closes, and no further work goes into these rules until the marks come back.

Mark	Means	What happens to the rule
T	TRUE — a real issue, worth raising with the supplier or the designer.	Kept as is.
F	FALSE — wrong, or not a real issue on this part.	Candidate for DELETION, not tuning. See below.
N	NOISE — technically true, but not worth an engineer's time.	Threshold raised, or the rule drops to advisory.

Why F is the column that matters. A false positive costs more trust than a missed finding does. An engineer who is shown one wrong finding stops reading the other nineteen. So a rule that collects an F is a candidate for deletion rather than tuning — we would rather ship twelve rules that are always right than nineteen that are usually right.

Please also fill in "what was missed". There is a box at the end of each part. Anything the tool should have caught and did not is worth more than any finding it did catch, because it becomes the next rule. A short finding list on a part is **not** a clean bill of health — each part states what the tool could not check, and that list is usually longer than the findings.

Part	Commodity	Manual cost	Features	Findings	Addressable £/part	Not checked
RH steering knuckle	casting (gravity)	~£16–18	172	4	—	4
Stub axle (PRCR002)	forging	~£30	128	2	—	5
25T servo horn	machining	~£2.2	105	2	£0.86	4
Front bumper	injection moulding	~£8–9	6	1	—	5
Seat LH cross-member	sheet metal	~£1.4	124	1	—	5
Fuel tank	blow moulding	~£20–30	905	2	—	5

2 of the 12 findings carry a £/part figure. The rest state, in place of a number, exactly which cost model is missing — a finding with no price says so rather than quietly showing zero. The £ figures are the cost of the feature as drawn at 50,000/yr, £60/hr machine and £25/hr labour; they are not a claim that the whole amount is recoverable.

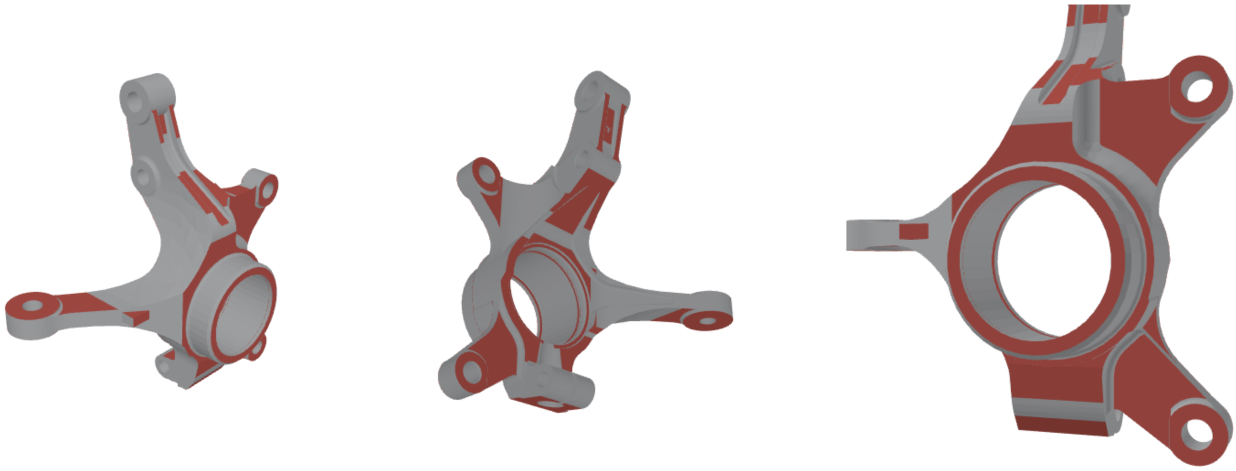
12 of 12 findings have an image, and 2 of those carry a note under the picture saying what it does or does not show — a feature on an internal surface, or one too small to see at part scale. Where an image is absent the reason is printed in its place: a part-level finding has no faces to highlight, which is different from a render that failed.

RH steering knuckle — casting (gravity)

File	Independent manual cost	Features measured	Findings	Addressable £/part
d8d13088-steering_knuckle_RH.stp	~£16–18	172	4	—

1. Wall below minimum draft for the casting route MAJOR

Instances	Measured	Threshold	£/part
60	draftDeg 0–0°	< 1°	—



Offending faces in red. Left: front isometric. Middle: rear isometric (the opposite side — half the findings on a real part sit where one view cannot see them). Right: close-up on the worst instance, aimed down the feature's own axis. 60 face(s): 1, 4, 9, 12, 18, 19, 22, 23, 31, 32, 33, 34, 46, 47, 48, 49, 53, 57, 60, 71 and more

Suggested fix: Add draft to at least 1° per side, or accept the ejection drag and the die wear it causes.

Source: ASM Handbook Vol. 15, Casting — permanent mould (gravity die) casting design

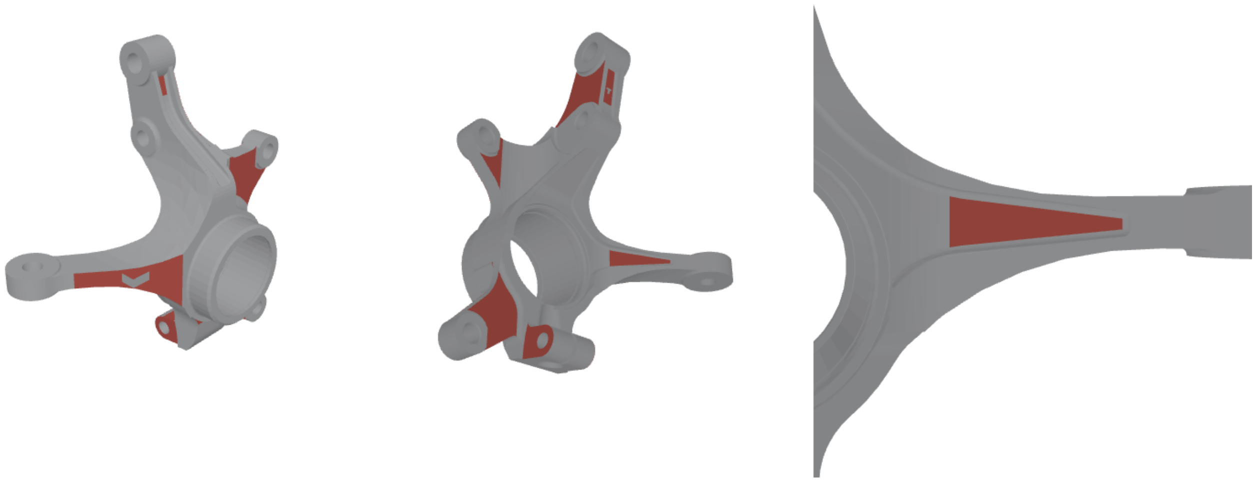
Not priced: Insufficient draft costs die wear and ejection drag. Converting that to £/part needs a die-life-versus-draft model, which this engine does not have.

`casting.draft.insufficient`

Mark	T		F		N		Comment:

2. Abrupt section change **MAJOR**

Instances	Measured	Threshold	£/part
18	sectionRatio 2.364–8.54:1	> 2:1	—



Offending faces in red. Left: front isometric. Middle: rear isometric (the opposite side — half the findings on a real part sit where one view cannot see them). Right: close-up on the worst instance, aimed down the feature's own axis. 18 face(s): 22, 23, 47, 48, 71, 129, 131, 132, 146, 179, 202, 210, 217, 218, 256, 257, 259, 289

Suggested fix: Blend the transition with a taper or a generous fillet so the section changes gradually; expect shrinkage porosity at the junction otherwise.

Source: ASM Handbook Vol. 15, Casting — design for solidification

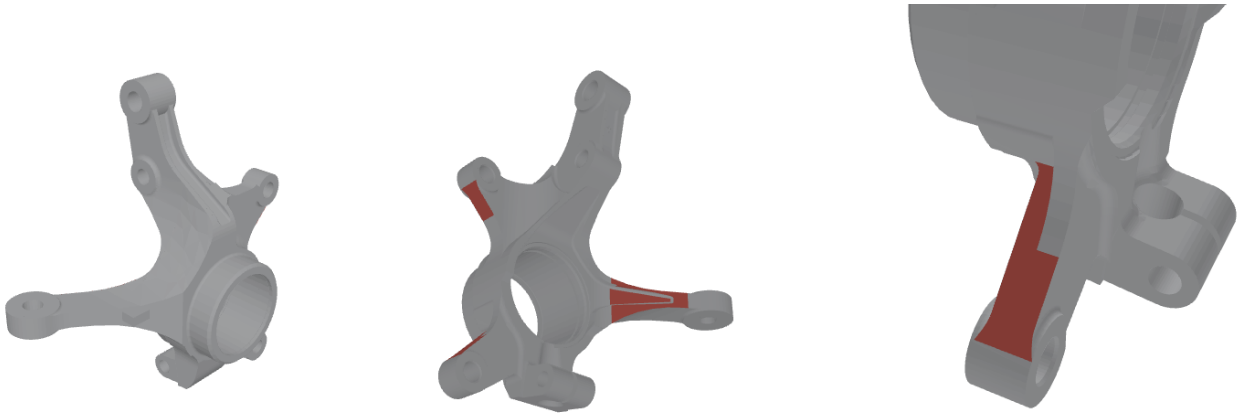
Not priced: Solidification shrinkage at a step change is a yield risk. No scrap-rate model exists, and inventing one would put a fabricated number next to measured ones.

`casting.section.abrupt-change`

Mark	T		F		N		Comment:

3. Undercut face — needs a slide or a core **MAJOR**

Instances	Measured	Threshold	£/part
6	angleToDrawDeg 97.37–135°	> 90°	—



Offending faces in red. Left: front isometric. Middle: rear isometric (the opposite side — half the findings on a real part sit where one view cannot see them). Right: close-up on the worst instance, aimed down the feature's own axis. 6 face(s): 61, 186, 241, 289, 295, 300

Suggested fix: Re-orient the parting line, or price a slide/core for this feature and carry the die cost and cycle penalty explicitly.

Source: NADCA Product Specification Standards for Die Castings — Undercuts and moving die components

Not priced: A slide or core adds die cost, but the casting die estimator does not price slides separately the way the injection-mould estimator does, so the delta is not derivable.

`casting.undercut.requires-core`

Mark	T		F		N		Comment:

4. Sharp internal corner MINOR

Instances	Measured	Threshold	£/part
14	radiusMm 0.25–0.25mm	< 1mm	—



Offending faces in red. Left: front isometric. Middle: rear isometric (the opposite side — half the findings on a real part sit where one view cannot see them). Right: close-up on the worst instance, aimed down the feature's own axis. 14 face(s): 89, 94, 96, 98, 100, 102, 106, 108, 110, 112, 114, 116, 118, 120

Note: The feature is very small at part scale — 168 of the mesh's 18,448 triangles carry it, so it reads as a few red pixels even zoomed in.

Suggested fix: Increase the fillet toward half the adjoining wall thickness to relieve the stress concentration and the local hot spot.

Source: ASM Handbook Vol. 15, Casting — fillets and corner radii

Not priced: A sharp corner is a stress raiser and a hot spot — a fatigue and yield risk, not a cost line.

`casting.fillet.sharp-internal-corner`

Mark	T		F		N		Comment:

What the tool could NOT check on this part

Printed so a short finding list above is not read as a clean part. Each line is a check the engine declared out of scope for this geometry, not a check that passed.

- Tolerance and datum callouts were not checked — they live on the drawing, not in the solid.
- Parting-line position was assumed to lie perpendicular to the draw direction; a different parting choice changes which faces are undercuts.
- Machining stock allowance on cast surfaces was not assessed — it is a process decision, not a measurable feature.
- Gating and feeding layout was not evaluated; hot spots are reported, but where to place risers is a foundry decision.

MISSED — what should this have caught and did not?

1.

2.

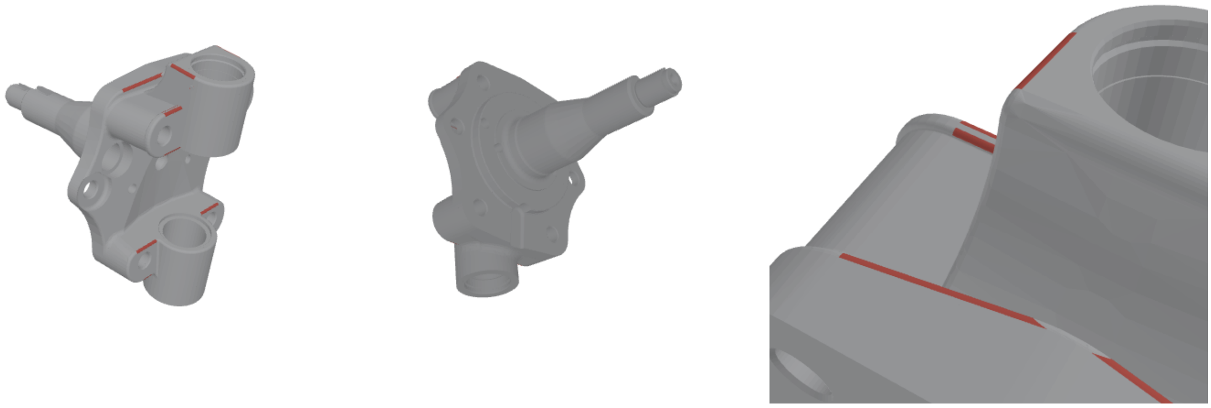
3.

Stub axle (PRCR002) — forging

File	Independent manual cost	Features measured	Findings	Addressable £/part
ee3d48dc-PRCR002.stp	~£30	128	2	—

1. Fillet too sharp for hot metal flow **MAJOR**

Instances	Measured	Threshold	£/part
18	radiusMm 1.5–2mm	< 3mm	—



Offending faces in red. Left: front isometric. Middle: rear isometric (the opposite side — half the findings on a real part sit where one view cannot see them). Right: close-up on the worst instance, aimed down the feature's own axis. 18 face(s): 34, 64, 69, 117, 119, 126, 169, 174, 177, 227, 279, 295, 296, 305, 308, 311, 313, 362

Suggested fix: Open the fillet to at least R3 mm. Hot metal will not turn a sharp corner: it folds instead, producing laps that only show up at magnetic-particle inspection, and the sharp die corner is where the die itself cracks first.

Source: ASM Handbook Vol. 14A, Metalworking: Bulk Forming — fillet and corner radii — Minimum fillet radius on hot forgings ≈ 3 mm

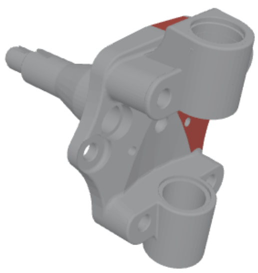
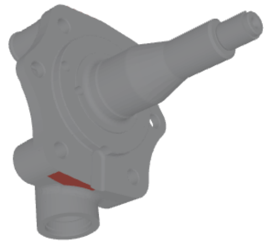
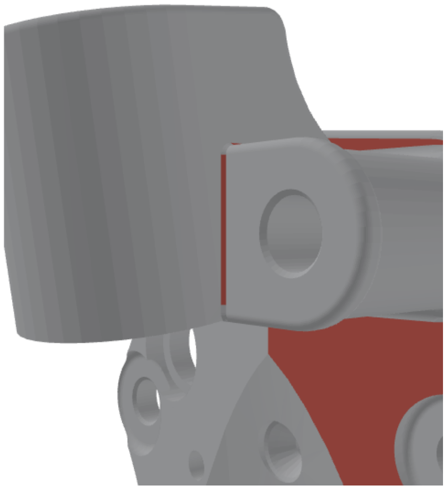
Not priced: Laps from folded metal are found at inspection. That is a scrap and rework risk with no modelled cost path.

`forging.fillet.too-sharp-for-metal-flow`

Mark T F N Comment:

2. Undercut — cannot release from an impression die MAJOR

Instances	Measured	Threshold	£/part
6	angleToDrawDeg 94.57–120°	> 90°	—

Offending faces in red. Left: front isometric. Middle: rear isometric (the opposite side — half the findings on a real part sit where one view cannot see them). Right: close-up on the worst instance, aimed down the feature's own axis. 6 face(s): 10, 11, 35, 172, 178, 194

Suggested fix: Move the parting line, or accept the feature as a machined operation after forging and carry that cost. Unlike casting or moulding, a forging die cannot buy its way out with a slide.

Source: ASM Handbook Vol. 14A, Metalworking: Bulk Forming — die design and parting

Not priced: The remedy is a post-forging machining operation, but the undercut is a FACE and the kernel measures no volume for what that operation would remove. Pricing it would mean inventing a cut. (An earlier version routed this to the hole pricer, which silently returned nothing because a face carries no diameter — a wrong mapping is worse than an honest gap.)

`forging.undercut.cannot-release`

Mark	T	☐	F	☐	N	☐	Comment:

What the tool could NOT check on this part

Printed so a short finding list above is not read as a clean part. Each line is a check the engine declared out of scope for this geometry, not a check that passed.

- Grain-flow alignment was not assessed. Whether the forged flow lines follow the principal load path is the single largest reason to forge a part rather than cast or machine it, and it depends on the die design and preform, not on the finished shape.
- Parting-line position was assumed perpendicular to the draw. A different parting changes which faces are undercuts and where the flash line falls; if the flash line crosses a sealing or highly-loaded face that is a defect the geometry cannot reveal.
- Machining stock allowance on forged surfaces was not checked — it is a process decision.
- Rib height-to-thickness was not computed: it needs rib identification, and the kernel does not yet separate a rib from any other thin planar region.
- Billet and preform design, and therefore realistic flash loss, were not evaluated.

MISSED — what should this have caught and did not?

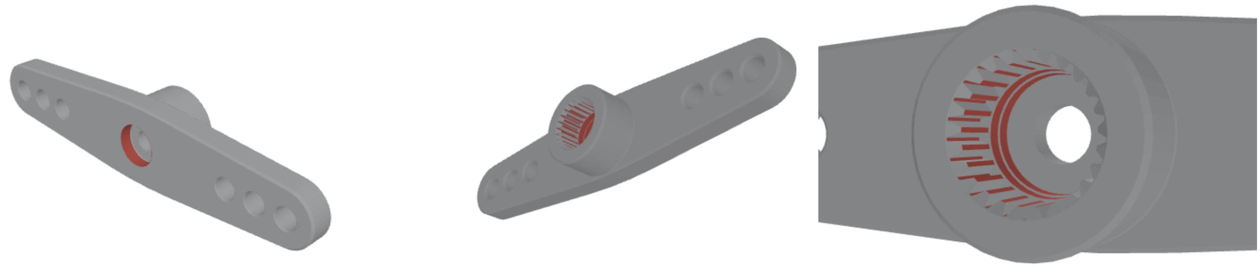
1. ______
2. ______
3. ______

25T servo horn — machining

File	Independent manual cost	Features measured	Findings	Addressable £/part
ed1e77f6-Aluminium_25T_Servo_Horn.step	~£2.2	105	2	£0.86

1. Non-preferred hole diameter — special tool ADVISORY

Instances	Measured	Threshold	£/part
4	diaMm 4.9–6.1mm	> 5mm	£0.55



Offending faces in red. Left: front isometric. Middle: rear isometric (the opposite side — half the findings on a real part sit where one view cannot see them). Right: close-up on the worst instance, aimed down the feature's own axis. 53 face(s): 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21 and more

Suggested fix: If the fit allows, move to ■5 mm and save a special tool or a separate bore/ream cycle.

Source: Machinery's Handbook — standard twist-drill diameters (metric series)

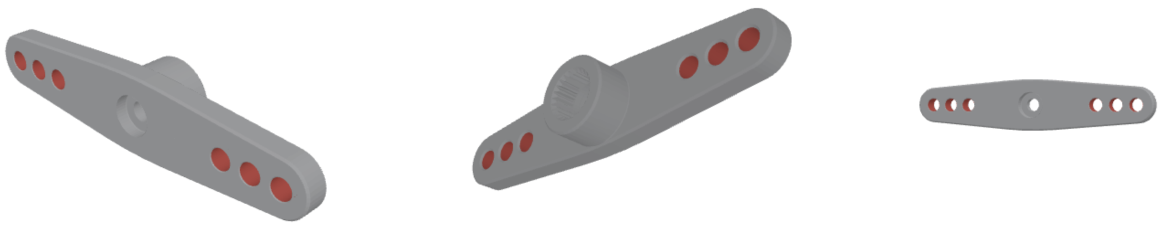
Cost basis: 0.07 min (featureMinutesEach, Ø5.1×1 mm) × £85.00/h machine+labour. Cost of the feature itself, not of the defect — the cycle model carries no peck-drilling penalty.

`machining.hole.non-preferred-diameter`

Mark	T		F		N		Comment:

2. Hole deeper than standard drill reach MINOR

Instances	Measured	Threshold	£/part
1	ldRatio 6.36:1	> 5:1	£0.31



Offending faces in red. Left: front isometric. Middle: rear isometric (the opposite side — half the findings on a real part sit where one view cannot see them). Right: close-up on the worst instance, aimed down the feature's own axis. 6 face(s): 112, 113, 114, 115, 116, 117

Suggested fix: Beyond ~5:1 needs peck-drilling or extended-reach tooling — slower cycle, higher tool cost and greater breakage risk. Open the diameter if the function allows.

Source: Machinery's Handbook — drilling: depth-to-diameter limits and standard drill sizes

Cost basis: 0.22 min (featureMinutesEach, Ø2.5×16 mm) × £85.00/h machine+labour. Cost of the feature itself, not of the defect — the cycle model carries no peck-drilling penalty.

`machining.hole.depth-beyond-standard-drill`

Mark **T** ☐ **F** ☐ **N** ☐ Comment:

What the tool could NOT check on this part

Printed so a short finding list above is not read as a clean part. Each line is a check the engine declared out of scope for this geometry, not a check that passed.

• Tolerance and surface-finish callouts were not checked — they are on the drawing, not in the solid, and they drive whether a feature is milled, ground or honed.
• Tool access and fixturing were not simulated: a feature can be geometrically fine and still unreachable in the chosen setup.
• Thread specifications were not verified against standard tap sizes.
• Workholding and part distortion under clamping were not assessed.

MISSED — what should this have caught and did not?

1.
2.
3.

Front bumper — injection moulding

File	Independent manual cost	Features measured	Findings	Addressable £/part
edd2f685-BUMPER.stp	~£8–9	6	1	—

1. Moulded wall below minimum draft MINOR

Instances	Measured	Threshold	£/part
2	draftDeg 0.04–0.04°	< 0.5°	—

Offending faces in red. Left: front isometric. Middle: rear isometric (the opposite side — half the findings on a real part sit where one view cannot see them). Right: close-up on the worst instance, aimed down the feature's own axis. 2 face(s): 218, 284
Note: Right-hand view is a CUT-AWAY: the material between the camera and the feature is removed, because the feature is on an internal surface. The feature is very small at part scale — 4 of the mesh's 24,427 triangles carry it, so it reads as a few red pixels even zoomed in.

Suggested fix: Add draft to at least 0.5° per side — more if the face is textured, roughly 1° per 0.025 mm of texture depth. Otherwise expect ejector drag marks.

Source: Injection-moulding design guidelines (widely published; e.g. Protolabs / Xometry design-for-moulding guides and the Plastics Design Library handbook) — Draft ≥ 0.5° per side on untextured walls; more with texture
Not priced: Ejector drag marks are a cosmetic reject risk, not a modelled cost.
`moulding.draft.insufficient`

Mark	T		F		N		Comment:

What the tool could NOT check on this part

Printed so a short finding list above is not read as a clean part. Each line is a check the engine declared out of scope for this geometry, not a check that passed.

- Draw direction is assumed to be +Z and is NOT derived from the part. 4 of 6 wall faces (67%) came out as undercuts, which no castable, mouldable or forgeable part can be — so the assumed draw is wrong for this shape and 4 undercut finding(s) were withdrawn rather than reported. Re-run with the correct parting direction, or read the draft findings below as provisional.
- Gate position and flow length were not evaluated — they are a tooling decision, not a feature of the solid.
- Weld-line position was not checked; whether a knit line falls on a functional or cosmetic face is a moulding-simulation question.
- Texture depth was not read, so the draft threshold applied is the untextured minimum — a textured face needs roughly 1° more per 0.025 mm of texture depth.
- Resin shrinkage and warp were not modelled; those need the material and a flow analysis.

MISSED — what should this have caught and did not?

1.

2.

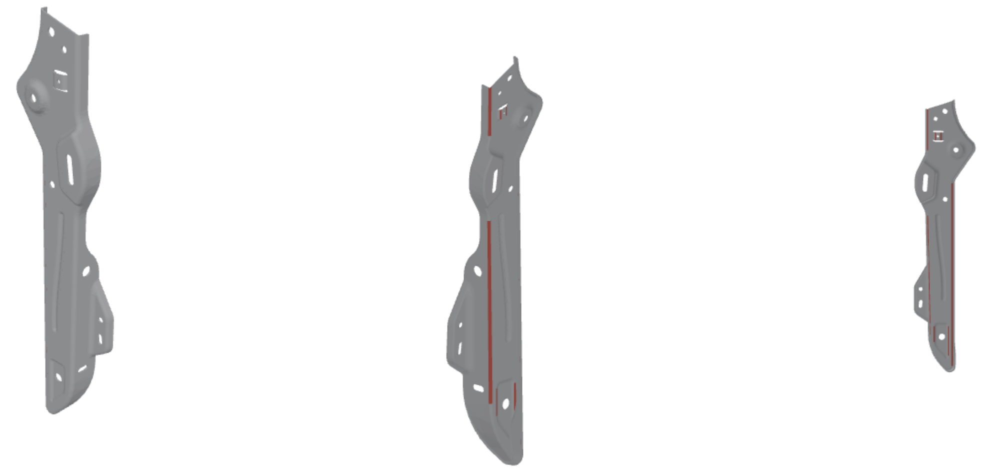
3.

Seat LH cross-member — sheet metal

File	Independent manual cost	Features measured	Findings	Addressable £/part
97fb714b-Seat_LH_Cross_Member.stp	~£1.4	124	1	—

1. Hole depth inconsistent with a sheet part ADVISORY

Instances	Measured	Threshold	£/part
2	ldRatio 9.537–20.546:1	> 3:1	—



Offending faces in red. Left: front isometric. Middle: rear isometric (the opposite side — half the findings on a real part sit where one view cannot see them). Right: close-up on the worst instance, aimed down the feature's own axis. 7 face(s): 80, 81, 224, 228, 233, 300, 304

Suggested fix: Confirm this is a formed collar / extruded hole rather than a punched one — the two cost very differently. If it is a plain hole, the part may not be sheet metal.

Source: Sheet-metal design guidelines (widely published; e.g. Xometry / Protolabs sheet-metal design guides, and Machinery's Handbook press-working sections) — Punched features in thin material

Not priced: This flags a possible mis-classification (formed collar versus punched hole), not a defect with a cost.

`sheetmetal.hole.deep-relative-to-diameter`

Mark T ☐ F ☐ N ☐ Comment:

What the tool could NOT check on this part

Printed so a short finding list above is not read as a clean part. Each line is a check the engine declared out of scope for this geometry, not a check that passed.

- Bend-to-hole distance was not checked: the kernel measures bend COUNT and total bend length but does not yet emit per-bend lines, so the distance from each hole to its nearest bend cannot be computed.
- Flange length against tooling minimum was not checked, for the same reason.
- Bend relief at notches was not checked — it needs per-bend endpoints.
- Bend radius against material thickness was not checked — it needs the per-bend inner radius.
- Blank utilisation was not checked — it needs a flat pattern, which the kernel does not unfold.

MISSED — what should this have caught and did not?

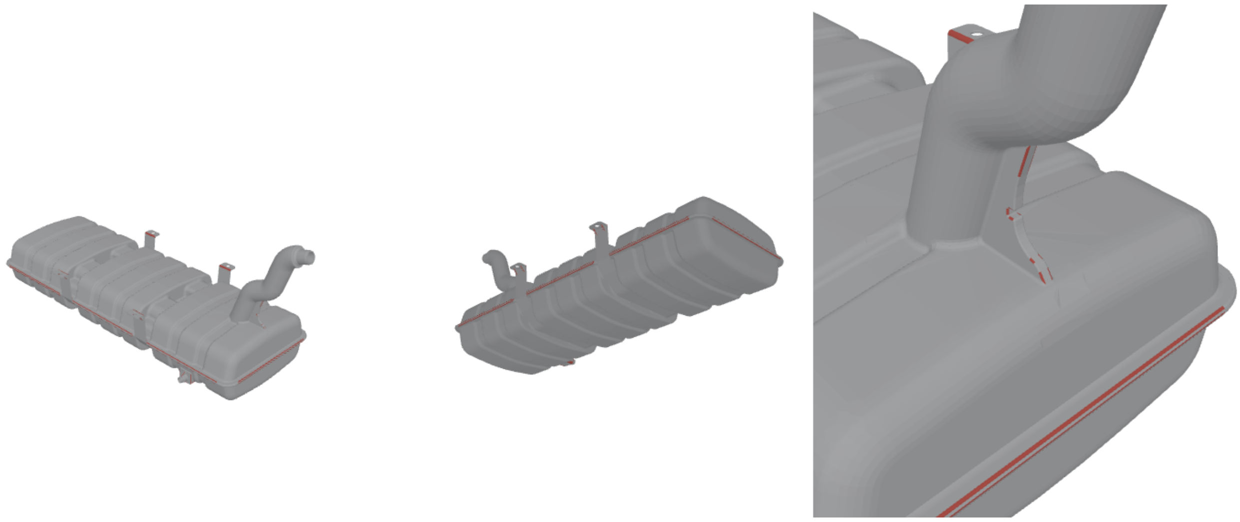
1.
2.
3.

Fuel tank — blow moulding

File	Independent manual cost	Features measured	Findings	Addressable £/part
b7a8495b-Fuel_tank.STEP	~£20–30	905	2	—

1. External corner too sharp for the parison to reach **MAJOR**

Instances	Measured	Threshold	£/part
104	radiusMm 1.5–8mm	< 8.91mm	—



Offending faces in red. Left: front isometric. Middle: rear isometric (the opposite side — half the findings on a real part sit where one view cannot see them). Right: close-up on the worst instance, aimed down the feature's own axis. 104 face(s): 920, 941, 949, 1619, 1621, 1623, 1626, 1628, 1632, 1641, 1644, 1646, 1650, 1653, 1658, 1661, 1667, 1676, 1678, 1680 and more

Suggested fix: Radius external corners to at least 2–3× wall. A corner is where the parison stretches hardest, so it thins to a fraction of nominal and cracks in service; sharp detail belongs on the least-stretched face.

Source: Blow-moulding design guidance (SPI/SPE extrusion blow-moulding design guides; mirrored in this tool's blow-advisor module) — External corner radius ≥ 2–3 × nominal wall

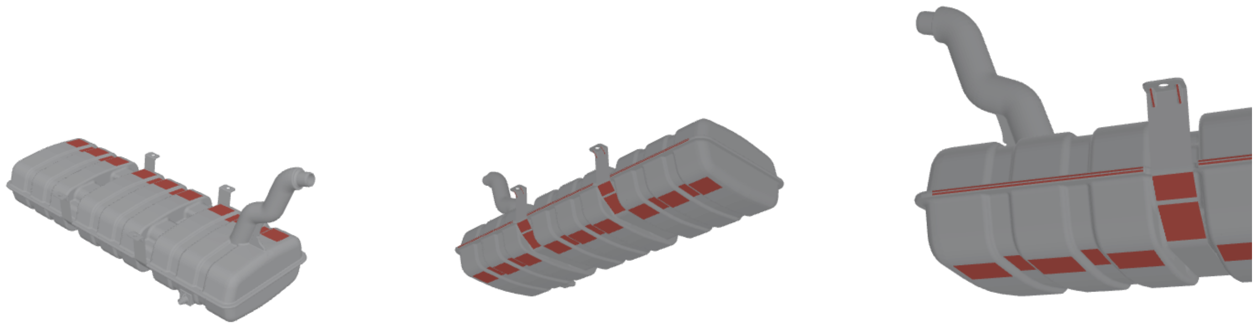
Not priced: Corner thinning is a burst and drop-test risk; pricing it needs a structural model.

`blow.corner.radius-below-2x-wall`

Mark	T	☐	F	☐	N	☐	Comment:

2. Undercut — the split mould cannot release it MAJOR

Instances	Measured	Threshold	£/part
84	angleToDrawDeg 96.37–161.57°	> 90°	—



Offending faces in red. Left: front isometric. Middle: rear isometric (the opposite side — half the findings on a real part sit where one view cannot see them). Right: close-up on the worst instance, aimed down the feature's own axis. 84 face(s): 58, 162, 329, 405, 423, 433, 452, 462, 497, 507, 530, 617, 635, 659, 677, 707, 709, 711, 976, 1128 and more

Suggested fix: Re-orient the split line, relax the feature so it draws, or price a moving insert. Deep undercuts on a blown part are often cheaper to trim after moulding than to tool for.

Source: Blow-moulding design guidance (SPI/SPE extrusion blow-moulding design guides; mirrored in this tool's blow-advisor module) — Undercuts and mould splits

Not priced: A third split or moving insert would be priced by a blow-mould estimator; this engine has a mould-cost model for injection tools only.

`blow.undercut.needs-split-or-insert`

Mark	T		F		N		Comment:

What the tool could NOT check on this part

Printed so a short finding list above is not read as a clean part. Each line is a check the engine declared out of scope for this geometry, not a check that passed.

- Wall figures are the characteristic 2·V/S thickness, not a per-face measurement. On a sealed thin shell the single-ray wall reading is unreliable, so local thin spots — which is where a blown part actually fails — cannot be located from this geometry.
- Blow-up ratio was not computed: it needs the parison die diameter, which is a process choice rather than a feature of the part.
- Pinch-off and weld-line position were not checked. On an extrusion-blown part the pinch weld is the weakest line and a common failure origin, and where it falls depends on the mould.
- Multi-layer barrier construction was not detected. An automotive fuel tank is typically a six-layer coextrusion, and costing it as a monolayer understates material significantly — the layer schedule has to be stated by an engineer.
- Top-load and drop performance were not assessed; those need the material and a structural check.

MISSED — what should this have caught and did not?

1.
2.
3.